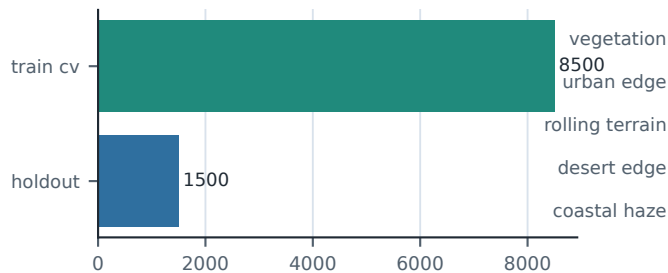
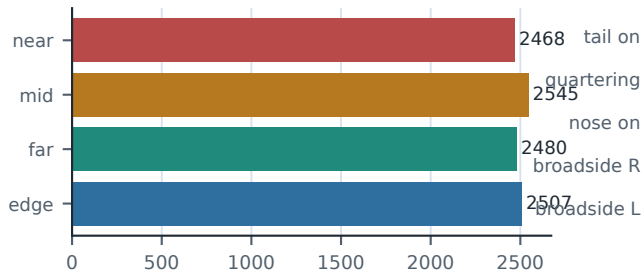


Scenario balance and leakage diagnostics

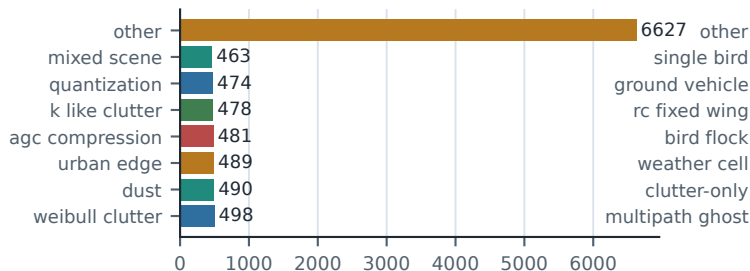
Split



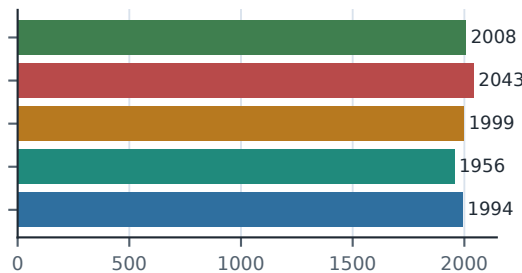
Range



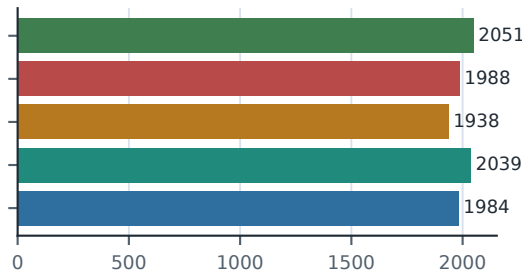
Noise



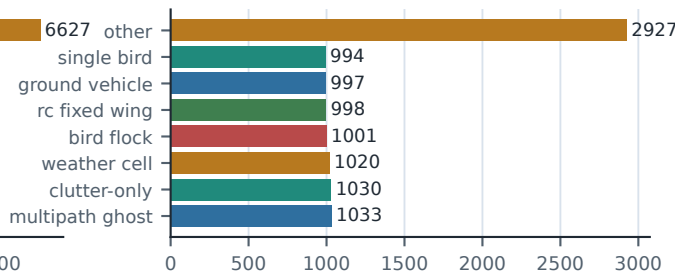
Site



Aspect



Hard-negative



Leakage checks

canary test: detected and blocked

stratum baseline AP: 0.005

metadata baseline AP: 0.004

label shuffle AP: 0.006

Marginal imbalance

aspect: 0.000

hard_negative_role: 0.032

noise: 0.000

range: 0.000

site: 0.000

split: 0.390

Scenario groups are locked before phase expansion. The paper evidence keeps label, site, range, aspect, noise, and hard-negative family counts together.